f(mn) = f(m) + f(n) − 1

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Let f be a function from $\mathbb{N}^*$ to $\mathbb{N}^*$ such that $f(mn) = f(m) + f(n) - 1$ for all integers m and n. Assume that $f(1980 \times 2002) = 11$ and that there are finitely many n with $f(n) = 1$.

Compute $f(16!)$.